Structured Problem Solving (SPS)

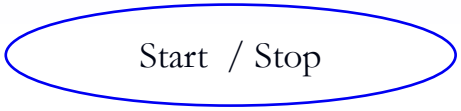
Implementation Model

Flow Charts

Implementation Model

Basic Flowchart symbols

Terminal



Flow line



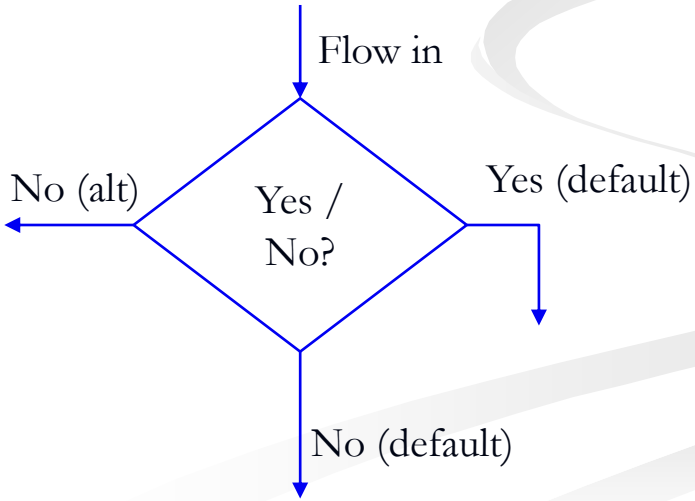
Input /Output



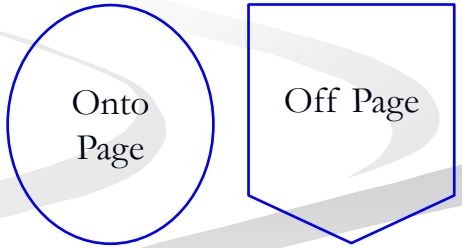
Process



Decision



Connectors



Implementation Model

Flowchart

Problem Statement:

"Develop a program that allows a user to enter a number, and that number is immediately displayed after it is entered."

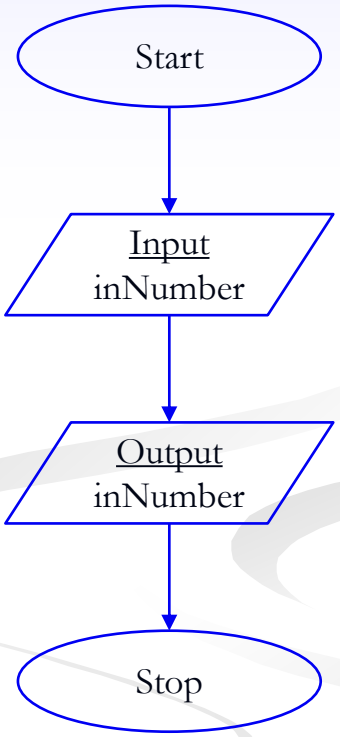
I/O Design:

Output:

- inNumber, (number entered)

Input:

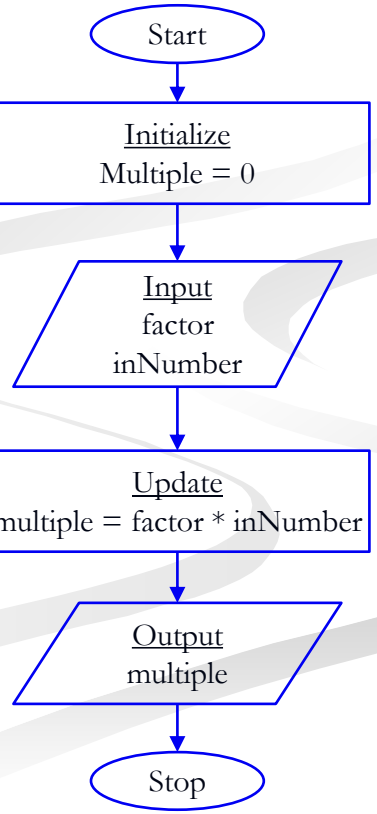
- inNumber, (number entered)



Essential and Implementation Models

Input single number and output a multiple

Problem Statement:	I/O Design:
"Develop a program that allows a user to enter a multiplicative factor and a number. The number multiplied by the factor is displayed immediately after the number is entered."	Output: ▪ multiple, (number multiplied by factor) Input: ▪ factor, (number entered as a multiplier) ▪ inNumber, (number entered)



Essential and Implementation Models

Fixed number of inputs and outputs

Problem Statement:

"Develop a program that allows a user to enter a multiplicative factor, and then to enter 4 numbers, one at a time. Each number multiplied by the factor is displayed immediately after the number is entered. After 4 numbers have been entered, the program ends."

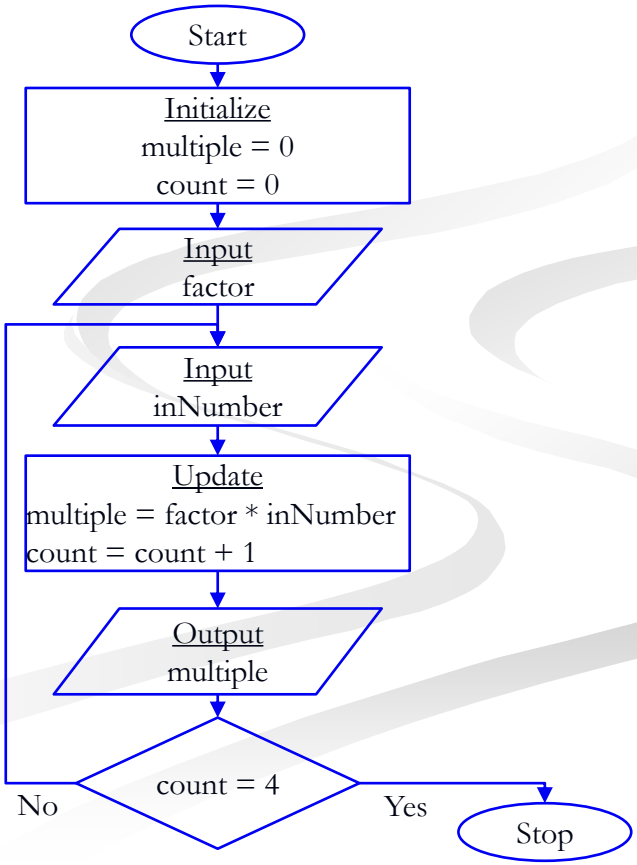
I/O Design:

Output:

- multiple, (number multiplied by factor)

Input:

- factor, (number entered as a multiplier)
- inNumber, (number entered)



Essential and Implementation Models

Inputs and outputs with a sentinel

Problem Statement:

"Develop a program that allows a user to enter a multiplicative factor, and then to enter numbers, one at a time. Each number multiplied by the factor is displayed immediately after the number is entered. The user enters numbers until the sentinel value of 999 is entered, and then the program ends."

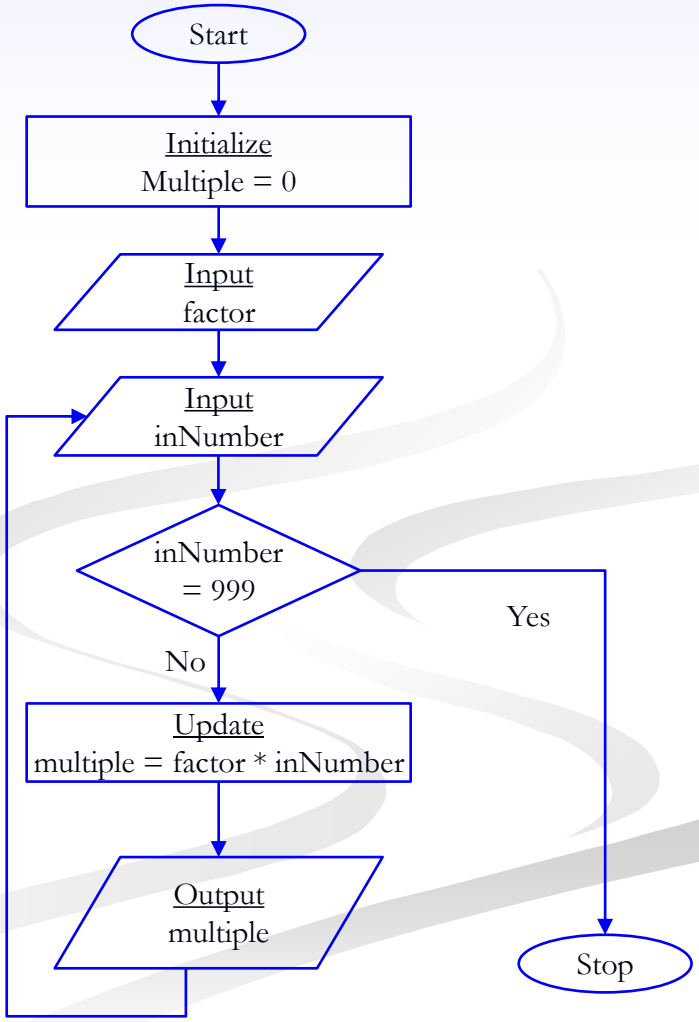
I/O Design:

Output:

- multiple, (number multiplied by factor)

Input:

- factor, (number entered as a multiplier)
- inNumber, (number entered)



Essential and Implementation Models

Cardinality, multiples, and sums

Problem Statement:

"Develop a program that allows a user to enter a multiplicative factor and a count of numbers that will be entered. The user then enters numbers, one at a time until the count is reached. The number multiplied by the factor is displayed immediately after the number is entered. After all numbers have been entered, the program displays separately the sum of all the numbers and the sum of all the multiples."

I/O Design:

Output:

- multiple, (number times the factor)
- sumNumbers, (sum of numbers entered)
- sumMultiples, (sum of multiples)

Input:

- factor, (number entered as a multiplier)
- card, (count of numbers to be entered)
- inNumber, (number entered)

